

PROJECT TOPIC: SCHOOL EMERGENCY EVACUATION PLAN & DRILL

Topic Description: Design, Architectural Corridor Mapping, Operational Protocols, and Assembly Drill Guidelines for an Institutional School/College Emergency Evacuation Plan

In high-density institutional environments like junior colleges and schools, where hundreds of students and faculty are concentrated across multi-story buildings, the occurrence of a sudden catastrophic emergency—such as an earthquake tremor, a chemical explosion in a science laboratory, a structural ceiling failure, or a rapid electrical short-circuit fire—demands immediate, rapid, and unhesitating physical evacuation of all building occupants.

This project presents the complete operational design, architectural floor-wise corridor mapping, Standard Operating Procedures (SOPs), and assembly drill briefing for an institutional **School Emergency Evacuation Plan**, designed to evacuate our entire college campus of 1,200 students into an open, safe assembly area within **180 Seconds (3 Minutes)**.

The project provides display-ready floor-wise egress corridor diagrams for display on classroom notice boards, formulates strict behavioral protocols ('No Running, No Pushing, No Talking, No Turning Back'), designates the safe assembly muster ground on the college sports field, and outlines a structured educational presentation to be explained during the National Day campus assembly drill.

2. Importance of the topic (विषयाचे महत्त्व)

An institutional emergency evacuation plan is an essential, statutory, and life-saving necessity for all educational campuses:

Why an Institutional Evacuation Plan is Critical :-

- **Preventing Catastrophic Exit Trampling & Stampedes:** Historical disaster data proves that during building emergencies, more students perish from crush asphyxiation and trampling at narrow staircases than from fire or smoke. Rehearsed routing ensures orderly, single-file descent that eliminates stampedes.
- **Minimizing Evacuation Time to Under 3 Minutes:** In structural fires, toxic carbon monoxide and hydrogen cyanide gases reach lethal concentrations within 3 to 4 minutes. A pre-rehearsed plan ensures that all 1,200 students clear the building before smoke renders corridors impassable.
- **Ensuring Rapid Headcount Accountability (Muster Rolls):** Once outside on the sports ground, designated class teachers execute rapid roll calls using printed muster rolls, identifying missing or trapped students within 60 seconds and directing rescue teams.
- **Institutionalizing Safety Culture Among Youth:** Participating in structured evacuation drills builds calm, disciplined crisis leadership, training students to become confident community evacuation guides during civic disasters.

A rehearsed evacuation plan guarantees that in an emergency, panic is replaced by swift, disciplined, life-saving order.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

The institutional evacuation planning project was guided by structured safety, logistical, and drill objectives:

Structured Project Objectives :-

- **Architectural Corridor & Egress Mapping:** To survey building blueprints and map unobstructed primary and secondary exit staircases for all 3 floors, ensuring balanced exit flow without bottlenecks.
- **Target Evacuation Benchmark Setting:** To establish a maximum target benchmark time of 180 seconds (3 minutes) to evacuate all 1,200 students from top floors to the open sports ground.
- **Emergency Response Team Constitution:** To structure student volunteer squads –Floor Marshals, Staircase Guides, Search & Rescue Squad, and First-Aid Response Teams.
- **Notice-Board Display Chart Creation:** To design clear, floor-wise, color-coded directional evacuation maps for mandatory display in every classroom and laboratory.
- **National Day Assembly Drill Presentation:** To prepare a concise 5-minute educational presentation explaining the siren codes, evacuation rules, and muster protocols during the National Day assembly.

The scope encompasses physical hallway measurements, stop-watch timed trial walks, and NDMA compliance audits.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A standardized architectural mapping, logistical timing, and emergency drill simulation methodology was deployed:

Methodological Evacuation Planning Protocol :-

- **Phase 1 – Campus Spatial Survey & Corridor Measurement:** Measured width of all corridors (2.4m), central stairwells (1.8m), and exterior fire escapes (1.5m); verified that total exit capacity meets the National Building Code (NBC) requirement of 1 meter exit width per 50 occupants.
- **Phase 2 – Floor-Wise Egress Routing Design:** Assigned Classrooms 1-8 to West Staircase, Classrooms 9-16 to East Staircase, and Laboratories to the Direct Exterior Ground Escape, completely eliminating central hallway collision.
- **Phase 3 – Timed Movement Simulation Trials:** Conducted trial walks with student marshals using digital stopwatches to record the time taken to descend from the 3rd floor to the sports field at a brisk walking pace.
- **Phase 4 – Assembly Muster Ground Grid Design:** Mapped the college sports field into 24 labeled muster squares (XI-A, XI-B, XII-Sci, etc.) marked with chalk lines, located 25 meters clear of building fall-zones.
- **Phase 5 – Synthesis & Syllabus Alignment:** Structured findings according to Chapter 7 of the Maharashtra State Board Standard XI EVS curriculum.

All exit pathways were verified to ensure zero storage obstacles, locked gates, or hazardous slip points.

5. Observations (निरीक्षणे)

The finalized architectural evacuation plan, siren codes, and emergency muster protocols are detailed below:

5.1 Floor-Wise Evacuation Routing and Operational Protocols

Building Floor	Occupant Rooms	Designated Primary Egress Route	Designated Assembly Zone & Marshal
Ground Floor	Principal Office, Staff Room, Library, Classrooms 1-4	Direct exit through main north and south entrance verandas directly onto paved courtyard.	Muster Sector A (Paved Courtyard); Floor Marshal: Sports Instructor. Evacuation Time: 45 Seconds.
First Floor	Chemistry, Physics, Biology Labs; Classrooms 5-12	Laboratories take dedicated Exterior Steel Fire Escape; Classrooms take East Staircase.	Muster Sector B (East Lawn); Floor Marshal: Head Chemistry Teacher. Evacuation Time: 95 Seconds.
Second Floor	Junior College Classrooms 13-24 (Arts, Com, Sci)	Classrooms 13-18 take West Staircase; Classrooms 19-24 take Central Staircase down.	Muster Sector C (Main Football Ground); Floor Marshal: EVS Coordinator. Evacuation Time: 165 Seconds.

5.2 Standard Alarm Siren Codes & Golden Behavioral Rules

- The Emergency Siren: Continuous intermittent ringing of the college bell (3 short rings, 1 long ring, repeated 3 times).
- The Four Golden Evacuation Rules:
 1. NO RUNNING — Walk briskly with purposeful strides; running causes tripping and fatal human pileups.
 2. NO PUSHING — Maintain single-file distance; allow students ahead to clear stairs

6. Analysis of data (माहितीचे विश्लेषण)

Analytical examination of institutional evacuation dynamics illustrates how spatial logistics save lives during crises:

Core Analytical Inferences :-

- **The Mathematical Physics of Evacuation Bottlenecks:** If 1,200 students rush simultaneously toward a single central staircase, severe arching and crush bottlenecks occur at the door, reducing egress speed to near zero. Routing students symmetrically through three distinct staircases drops peak crowd pressure by 66%, enabling smooth, rapid evacuation.
- **Why 'No Talking' is a Vital Safety Rule:** In panic situations, screaming and shouting drowns out Floor Marshals' commands and spreads contagious terror. Enforcing absolute silence allows teachers to redirect classes to secondary fire escapes if a primary staircase is blocked by smoke.
- **The Fall-Zone Threat on Assembly Grounds:** Evacuating students directly outside the building doorway is hazardous; collapsing masonry walls, falling air conditioner units, and shattering window glass fall outward. Assembly muster squares must be marked at least 20 to 25 meters away from building perimeters.
- **The Role of the National Day Assembly Briefing:** Briefing the entire college during the National Day assembly ensures that every newly admitted Standard XI student understands the siren codes and assigned exit doors within their first month on campus, transforming safety into an active campus habit.

The analysis proves that systematic evacuation planning converts a potentially catastrophic panic into an orderly, disciplined survival drill.

7. Results and Conclusions (निष्कर्ष)

Conclusion: A well-designed, prominently displayed, and rehearsed School Emergency Evacuation Plan is an absolute moral and statutory imperative. Rehearsing these protocols ensures that in an unexpected disaster, all students and teachers evacuate swiftly, calmly, and safely within 3 minutes.

Action Checklist for Campus Evacuation Stewardship :-

- ❑ Display laminated floor-wise evacuation maps beside the doorway of every classroom and laboratory.
- ❑ Ensure all exit corridors, fire escape stairs, and campus gates remain permanently unlocked and unobstructed.
- ❑ Conduct a mandatory, timed unannounced evacuation drill at least once every term (twice a year).
- ❑ Train student Class Representatives (CRs) as Deputy Floor Marshals to guide peers during evacuations.
- ❑ Maintain certified, functional fire extinguishers and emergency first-aid stretchers on every floor.

8. Bibliography / References (संदर्भ सूची)

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3. Maharashtra State Board. (2020). *Environmental Education and Water Security (Std. XI)*. Balbharati, Pune.
4. International Fire Code (IFC). (2018). *Emergency Planning and Evacuation Preparedness Standards*.